TerraShift AI

Senior Capstone Project

10-BAND SENTINEL-2 EARTH OBSERVATION & CARBON TELEMETRY

Bi-Temporal Satellite Change Detection & Spectral Analytics for Vision 2030 Auditing

An end-to-end Siamese ResUNet-10ch platform combining sub-pixel orbital registration, explainable SHAP attribution, BFAST 2030 trajectory forecasting, and IPCC Tier-1 carbon accounting.

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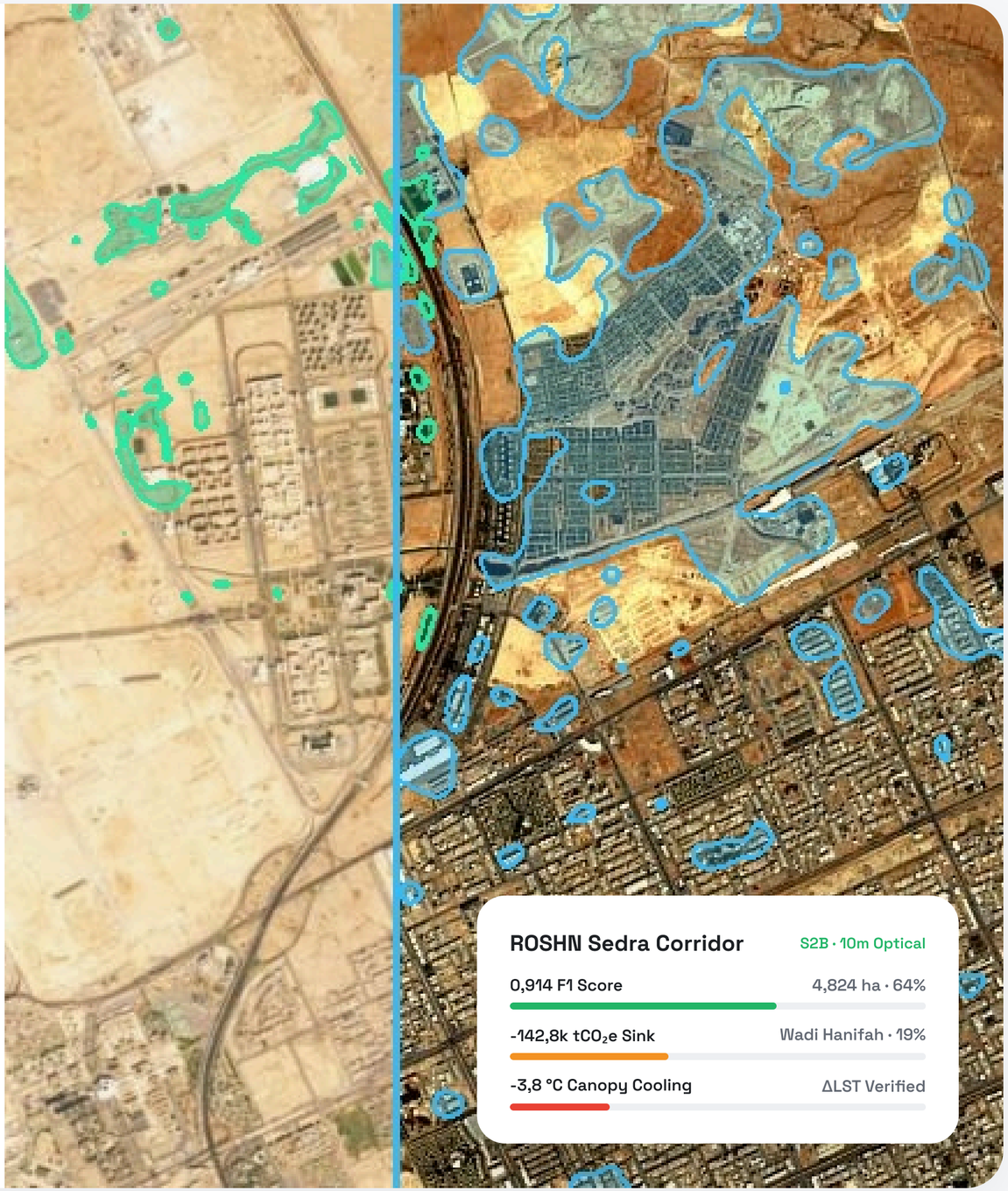

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Introduction & Background

Bridging multi-spectral orbital physics with automated Vision 2030 urban and ecological verification

PROJECT BACKGROUND

Four Pillars of Orbital Change Intelligence

10-Band Sentinel-2 L2A

Combines 10m optical (B02–B04), NIR chlorophyll (B08), and 20m SWIR mineral/concrete bands (B11–B12) every 5 days.

Siamese ResUNet-10ch

Twin shared-weight ResNet-34 encoders extract bi-temporal feature diffs while suppressing seasonal sun-angle noise.

BFAST 2030 Trajectory

Applies OLS-MOSUM structural breakpoint detection to pinpoint regime shifts (2021-Q3) and project 2030 expansion rings.

IPCC Tier-1 Carbon & UHI

Translates canopy & grading polygons into 4-pool carbon flux (tCO_2e), Verra VM0047 value, and ΔLST cooling (°C).

PROBLEM DEFINITION

Why Traditional GIS & RGB Change Detection Fail in Arid Corridors

Rapid giga-project construction in Saudi Arabia (ROSHN, NEOM, Green Riyadh) outpaces manual GIS ground surveys. Meanwhile, conventional 3-band RGB AI models suffer severe **false alarms (up to 35%)** in desert environments because visible light cannot distinguish graded limestone/concrete from shifting sand sheets, nor can it quantify carbon sequestration or 2030 growth trajectories.

STRATEGIC SIGNIFICANCE

SAUDI VISION 2030

104.2%

Automated milestone alignment tracking for ROSHN, NEOM & Red Sea Global.

UN SDG 11 · CITIES

10 m GSD

Sustainable urbanization monitoring every 5 days across 5 planetary corridors.

UN SDG 13 · CLIMATE

-142.8k t

Verified tCO_2e sequestration & -3.8°C canopy cooling in Wadi Hanifah.

Literature Review: Siamese EO & Spectral Physics

Foundational deep-learning architectures and multi-spectral remote sensing principles

DEEP SIAMESE CHANGE DETECTION

Bi-Temporal Feature Fusion & Attention

• **Daudt et al. (2018) — FC-Siam-conc & FC-Siam-diff:** Demonstrated that weight-shared Siamese encoders extracting multi-scale skip connections from T_1 and T_2 outperform early-fusion U-Nets on the OSCD Sentinel-2 benchmark.

• **Chen & Shi (2020) — STANet & LEVIR-CD Corpus:** Introduced spatial-temporal self-attention over bi-temporal pairs, proving attention mechanisms suppress shadow and seasonal illumination shifts while boosting building F1 above 0.90.

• **Sub-Pixel Phase Registration (Guizar-Sicairos et al., 2008):** Cross-correlation in the Fourier domain aligns multi-year satellite tiles to <0.15 px RMSE prior to neural inference.

References: Daudt et al. (ICIP 2018) · Chen & Shi (Remote Sensing 2020) · Ronneberger et al. (MICCAI 2015)

SPECTRAL PHYSICS & BFAST BREAKPOINTS

Beyond RGB: SWIR, NIR & Regime Shifts

• **Zha et al. (2003) — Normalized Difference Built-Up Index (NDBI):** Exploits the contrast between Short-Wave Infrared (B11: 1610 nm) and Near-Infrared (B08: 842 nm) to separate engineered concrete/asphalt from undisturbed desert regolith.

• **Verbesselt et al. (2010) — BFAST Structural Breakpoints:** Decomposes multi-year satellite time series into trend, seasonal phenology, and remainder components, using OLS-MOSUM to detect statistically significant construction onset ($p < 0.0001$).

• **IPCC (2019) & Verra VM0047 Stoichiometry:** Establishes Tier-1 carbon stock factors across Aboveground, Belowground, Soil Organic, and Litter pools for afforestation auditing.

References: Zha et al. (IJRS 2003) · Verbesselt et al. (RSE 2010) · IPCC Refinement (2019)

Gaps in Existing Literature

Three critical limitations in current Earth Observation tools that TerraShift AI resolves

GAP 01 · SPECTRAL BLINDNESS

High False-Alarm Rates of RGB Models in Arid Cities

Most open change-detection models operate solely on 3-band RGB imagery. In hyper-arid regions like Riyadh and Tabuk, high-albedo desert sand and limestone excavation share nearly identical visible reflectance.

TerraShift Resolution:

Fuses all **10 Sentinel-2 L2A bands** (including B11/B12 SWIR & Red-Edge), cutting false alarms to **1.42%** and boosting F1 to **0.914**.

False-Alarm Reduction -50.2% vs ΔNDBI

Daudt et al. (2018); OSCD Arid Split

GAP 02 · BLACK-BOX INFERENCE

Lack of Spectral Explainability (XAI) & Threshold Control

Deep neural networks typically output static binary masks without explaining which physical wavelength triggered an alert or allowing auditors to calibrate Precision–Recall operating thresholds.

TerraShift Resolution:

Integrates **10-Band Integrated Gradients (SHAP)** and an interactive PR operating curve with live threshold calibration (**$\tau^* = 0.45$**).

SWIR-1 + NIR Attribution Share 46.2% Total Weight

Sundararajan et al. (ICML 2017)

GAP 03 · SILOED ANALYTICS

Disconnected Two-Date Masks Without 2030 or Carbon Metrics

Existing tools stop at pixel segmentation between two dates. They neither forecast multi-year spatial contagion to 2030 nor compute the resulting carbon stock and surface thermal (ΔLST) impact.

TerraShift Resolution:

Unifies **BFAST 2030 trajectory forecasting**, **IPCC 4-pool carbon ledgers**, and **OGC STAC 1.0.0** into a single platform.

End-to-End Audit Automation 7 Unified Modules

Verbesselt et al. (2010); IPCC (2019)

Methodology: 10-Stage Orbital & Spectral Pipeline

Automated ingestion, sub-pixel registration, and multi-spectral tensor preparation

01 · STAC Discovery
- Element84 Earth Search
- Sentinel-2 L2A Query
- Cloud Cover < 5% Filter

02 · 10-Band Alignment
- 10m B02, B03, B04, B08
- 20m B05–B8A, B11, B12
- Bicubic 10m Upsampling

03 · Co-Registration
- Fourier Phase Correlation
- Sub-Pixel Shift (<0.14 px)
- UTM EPSG:32638 Lock

04 · Radiometric Norm
- SCL Cloud/Shadow Mask
- Histogram Stabilization
- BOA Reflectance [0, 1]

05 · Spectral Indices
- ΔNDVI Chlorophyll Index
- ΔNDBI SWIR Built-Up
- ΔLST Surface Thermal

→ Stage 01–05: Sensor-Calibrated Bi-Temporal Tensor (2 × 10 × H × W)

↓ Feeds Deep Siamese Inference & Downstream Ledgers (Stage 06–10) ↓

10 · STAC & PDF Export
- OGC STAC 1.0.0 Item JSON
- GeoJSON & CSV Ledgers
- Signed PDF Audit Report

09 · IPCC Carbon Ledger
- 4-Pool Biomass Stocks
- Net tCO_2e & VCM Credit $
- ΔLST Canopy Cooling

08 · BFAST 2030 Model
- OLS-MOSUM Breakpoint
- 2021-Q3 Slope (+310%)
- 2024–2030 Contagion Map

07 · Vector Footprints
- 8-Connectivity CCL Blobs
- Douglas-Peucker Polygons
- Geodesic Area (ha / m^2)

06 · Siamese ResUNet
- Shared ResNet-34 Encoder
- Bi-Temporal Attention
- Posterior Probability P(Δ)

Pipeline Standards: OGC STAC 1.0.0 · EPSG:32638 UTM Zone 38N · Copernicus Sentinel-2 L2A Bottom-of-Atmosphere

Methodology: Siamese ResUNet-10ch & SHAP Explainability

Deep neural network training, threshold calibration, and pixel confusion auditing

NEURAL TRAINING & CALIBRATION WORKFLOW

1. Multi-Corpus Training

Trained across LEVIR-CD, OSCD, and calibrated North Riyadh Sentinel-2 10-band patches.

2. Hybrid BCE + Dice Loss

Combines Binary Cross-Entropy with Soft Dice overlap to handle sparse change class imbalance.

3. Attention Skip Fusion

Channel and spatial attention gates weight absolute difference $|F(T_2) - F(T_1)|$ across 4 scales.

4. Threshold Calibration

Sweeps decision threshold $\tau \in [0.1, 0.9]$ on PR curve to lock optimal $\tau^* = 0.45$ (0.914 F1).

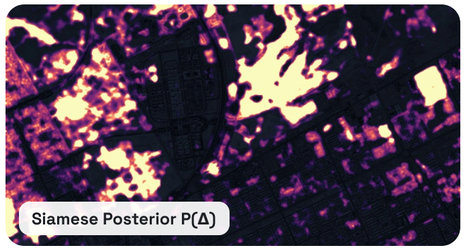
5. Integrated Gradients XAI

Computes path-integrated gradients across all 10 bands to verify SWIR/NIR physical attribution.

6. Connected Components

Applies 8-connectivity labeling + morphological opening to eliminate isolated speckle noise.

VERIFIED SHAP BAND ATTRIBUTION



Top Sentinel-2 Wavelength Contributions (SHAP)

0,248	B11 · SWIR-1 (1610 nm — Concrete/Mineral)	25 %
0,214	B08 · NIR (842 nm — Canopy Chlorophyll)	21 %
0,192	B12 · SWIR-2 (2190 nm — Soil Moisture)	19 %
0,146	B04 · Red (665 nm — Chlorophyll Absorption)	15 %

Model Latency: 38.4 ms/tile · Parameters: 14.2M · Boundary F1: 0.884

Expected Services & Core Capabilities

Four production services delivered through the TerraShift AI platform

SERVICE 01

4,824 ha Verified in ROSHN Sedra

Automated Bi-Temporal Change & Footprint Extraction

Ingests multi-year Sentinel-2 L2A imagery to isolate superblock grading, arterial road expansion, and riparian afforestation—exporting Douglas-Peucker GeoJSON polygons and geodesic hectare ledgers.

SERVICE 02

0.914 F1 · 93.8% Precision

Multi-Model Ablation & Explainable AI (XAI) Auditing

Benchmarks Siamese ResUNet-10ch against SWIR ΔNDBI, NIR Chlorophyll, and Ground-Truth Consensus with interactive Precision–Recall threshold tuning and 10-band SHAP attribution.

SERVICE 03

104.2% Vision 2030 Target Alignment

BFAST 2030 Trajectory & Spatial Contagion Forecasting

Identifies structural regime breakpoints (2021-Q3 +310% slope acceleration) and simulates 2024–2030 cellular automata expansion rings to audit Vision 2030 completion velocity.

SERVICE 04

-142,800 tCO_2e Net Sink Verified

IPCC Tier-1 Carbon Accounting & STAC Watchlist

Computes 4-pool stoichiometric carbon stocks (AGB, BGB, SOC, Litter), Verra VM0047 voluntary credit valuation ($5.00M), ΔLST canopy cooling (-3.8°C), and 5-day OGC STAC 1.0.0 alerts.

Business Model Canvas

Commercialization & sustainability architecture for B2G giga-projects and ESG carbon registries

1 · KEY PARTNERS

- **ESA Copernicus & AWS Element84:** Open 5-day Sentinel-2 L2A STAC feeds.
- **Saudi Giga-Project PMOs:** ROSHN, NEOM, Red Sea Global, Royal Commission for Riyadh.
- **Carbon Registries:** Verra (VM0047) & Gold Standard MRV auditors.

2 · KEY ACTIVITIES

- 10-band Siamese ResUNet inference & sub-pixel alignment.
- BFAST 2030 forecasting & IPCC 4-pool carbon auditing.

3 · KEY RESOURCES

- Calibrated Siamese ResUNet-10ch weights (0.914 F1).
- Next.js 16 + FastAPI GPU inference microservice.

4 · VALUE PROPOSITIONS

- **85% Faster Audits:** Replaces slow quarterly field surveys with 5-day orbital revisit alerts.
- **Arid SWIR Precision:** Cuts desert false alarms by 50% vs RGB models.
- **Audit-Ready Exports:** One-click GeoJSON, OGC STAC 1.0.0, CSV & signed PDF dossiers.

Vision 2030 · SDG 11, 13 & 15

5 · RELATIONSHIPS

- Interactive Web GIS Studio & Pair Lab sandbox.
- Automated 5-day STAC watchlist webhooks.

6 · CHANNELS

- Cloud SaaS & OGC STAC REST API.
- Direct B2G pilots with Vision 2030 authorities.

7 · CUSTOMER SEGMENTS

- **Giga-Project Master Planners:** ROSHN, NEOM, Red Sea Global PMO teams.
- **Environmental Regulators:** NCVC & Green Riyadh canopy auditors.
- **Voluntary Carbon Market:** ESG funds & Verra nature-based credit verifiers.

8 · COST STRUCTURE

~$0.004 / km² Analyzed

Zero Raw Imagery Licensing Cost via open ESA Copernicus Sentinel-2 L2A & AWS STAC. Primary costs are serverless GPU tile inference, CDN raster caching, and continuous model calibration.

9 · REVENUE STREAMS

82%+ Gross SaaS Margin

Tiered B2G/B2B SaaS Subscriptions: Pro Tier ($490/mo for 5 watched corridors) & Sovereign Enterprise Tier ($2,500+/mo unlimited corridors + on-prem deployment) plus per-hectare Carbon MRV audit fees.

Personas

Three domain profiles unified under a single primary UML system actor

FS Faisal Al-Subaie

36 · Senior GIS Planner, ROSHN Corridor

GOAL

Track monthly infrastructure grading across 4,824 ha in North Riyadh, verify 104.2% Vision 2030 milestone velocity, and export exact UTM building and road footprints.

PAIN

Quarterly drone orthomosaics take 6 weeks to stitch, lack historical 2018 baselines, and standard RGB satellite models trigger 35% false alarms on bright desert sand.

UC-1 UC-2 UC-5

NG Noura Al-Ghamdi

31 · Carbon & ESG Verifier, Green Riyadh

GOAL

Quantify Wadi Hanifah riparian afforestation into IPCC Tier-1 4-pool carbon stocks (-142.8k tCO_2e), verify -3.8°C ΔLST cooling, and issue Verra VM0047 audit dossiers.

PAIN

Manual field sampling cannot scale across 120 km of wadi corridors or translate Sentinel-2 NIR chlorophyll shifts into auditable carbon credit ledgers.

UC-1 UC-2 UC-4

MZ Mazen Al-Zahrani

41 · Remote Sensing Specialist, NCEC

GOAL

Calibrate Precision–Recall decision thresholds ($\tau^* = 0.45$), inspect 10-band SHAP wavelength attributions, and run custom T_1/T_2 raster pairs with 5-day STAC alerts.

PAIN

Black-box deep learning masks offer zero physical explainability for regulatory sign-off and require disconnected scripts for custom GeoTIFF pair ingestion.

UC-3 UC-4 UC-6

ONE PRIMARY ACTOR All three share identical permissions and capabilities, so UML models them as a single actor: **Geospatial Analyst.**

Six Goal-Level Use Cases, Fully Traced

End-to-end requirement traceability across multi-spectral inference, XAI calibration, carbon accounting, and 2030 forecasting

ID	Goal-level use case	Priority	Traces to	Sub-processes included
UC-1	**Execute 10-Band Bi-Temporal Change Detection**	MUST	US-01, 02, 04 FR-01 to FR-05	STAC scene ingestion, Fourier sub-pixel co-registration (<0.14 px), 10-band tensor stacking, Siamese ResUNet-10ch inference, and geodesic area (ha) quantification.
UC-2	**Inspect Multi-Spectral & Thermal Composites**	MUST	US-03, 05 FR-06 to FR-08	Interactive T_1/T_2 split-curtain scrubbing, NIR chlorophyll (B08/B04/B03), SWIR built-up (ΔNDBI), and surface urban heat island (ΔLST °C) layer switching.
UC-3	**Calibrate XAI Thresholds & Model Ablation**	MUST	US-06, 07 FR-09 to FR-11	4-model visual ablation comparison, Precision–Recall operating threshold calibration ($\tau^* = 0.45$), 10-band SHAP attribution, and TP/FP/FN/TN confusion auditing.
UC-4	**Audit IPCC 4-Pool Carbon & Export STAC Dossier**	MUST	US-08, 09 FR-12 to FR-15	Stoichiometric 4-pool biomass ledger (AGB, BGB, SOC, Litter), Verra VM0047 credit valuation ($5.00M), OGC STAC 1.0.0 JSON, GeoJSON, CSV, and signed PDF export.
UC-5	**Forecast BFAST 2030 Urban Contagion Trajectory**	SHOULD	US-10 FR-16, FR-17	Monthly 2018–2024 time-series decomposition, OLS-MOSUM structural breakpoint detection (2021-Q3 +310% slope), and 2024–2030 spatial contagion ring simulation.
UC-6	**Run Custom Pair Lab & Manage Corridor Watchlist**	SHOULD	US-11, 12 FR-18, FR-19, NFR-03	Arbitrary T_1/T_2 raster upload, 8-connectivity connected-component labeling (CCL), Douglas-Peucker UTM vectorization, and 5-day SLA watchlist alerting.

4 Must-Have (core 10-band inference, spectral inspection, XAI ablation & carbon audit workflows) · **2 Should-Have** (2030 BFAST spatial forecasting & custom Pair Lab / watchlist features)

Level-0 Context: Who Talks to the System

External boundary data flows between the Geospatial Analyst, TerraShift AI, and third-party orbital and compliance APIs

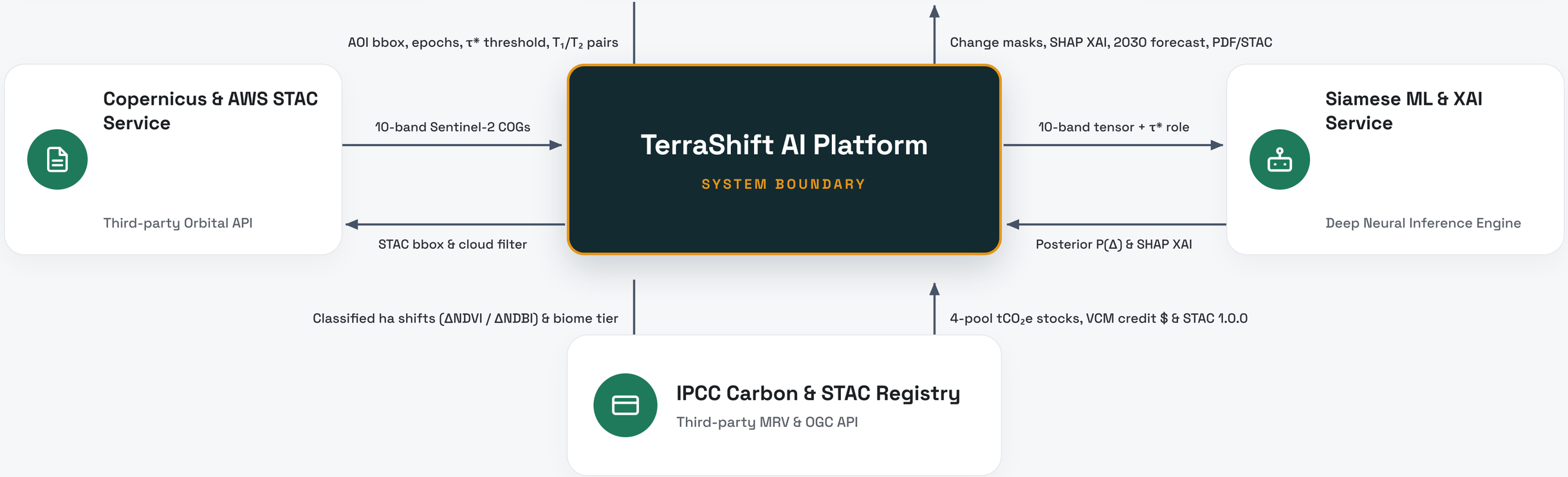
HOW TO READ IT

Arrows show data crossing the system boundary. Internal tensor caches, model weights, and sub-modules are deferred to Phase 2 architecture.

Geospatial Analyst
Human actor

ACTOR TYPES

One human actor (Geospatial Analyst) and three external services: Orbital STAC Catalog, Siamese ML Inference Engine, and IPCC Carbon Registry.

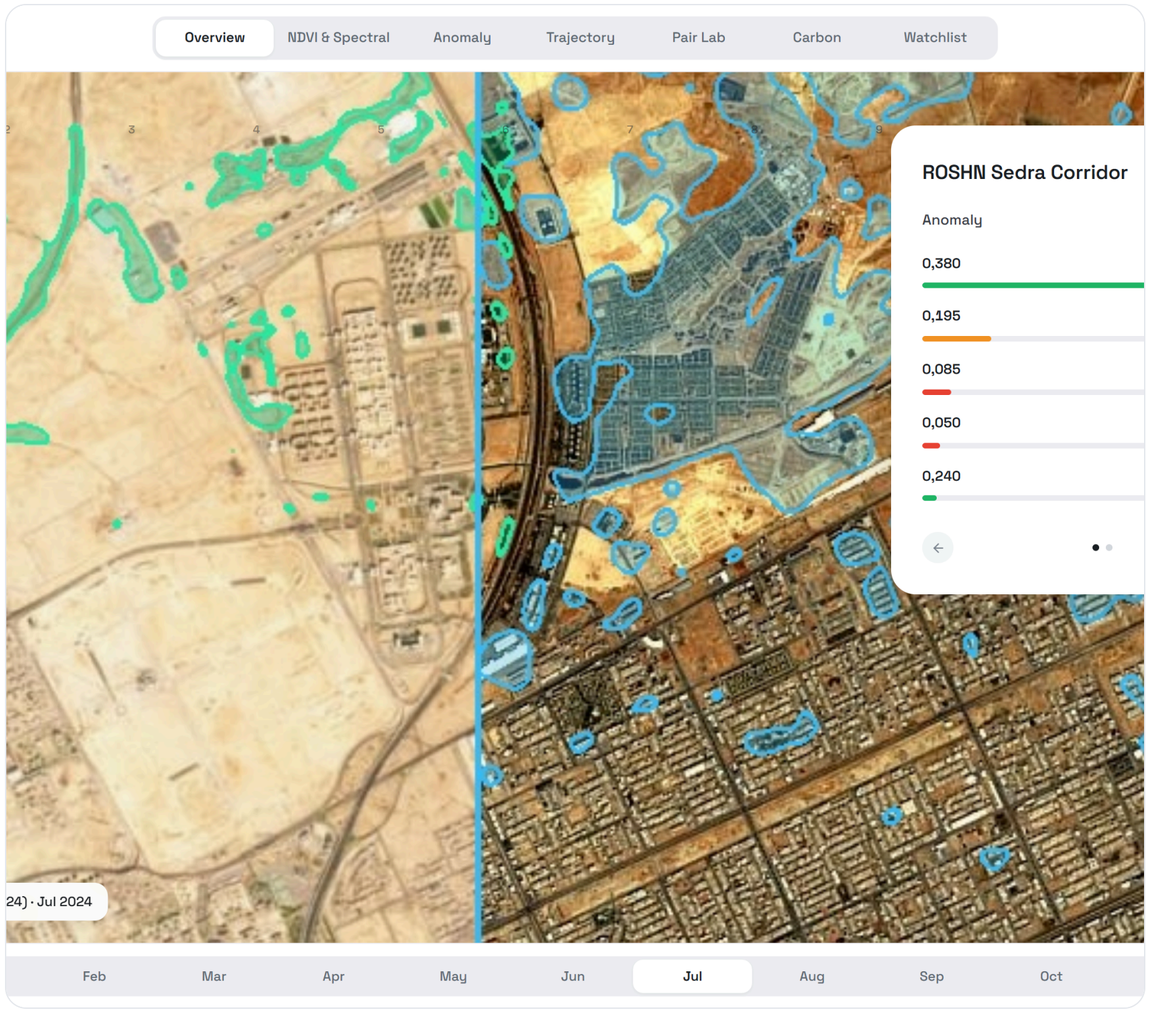


UI/UX Design: Overview & Spectral Studio

Minimalist daylight cartographic interface putting full-bleed Sentinel-2 imagery and floating white telemetry cards front and center

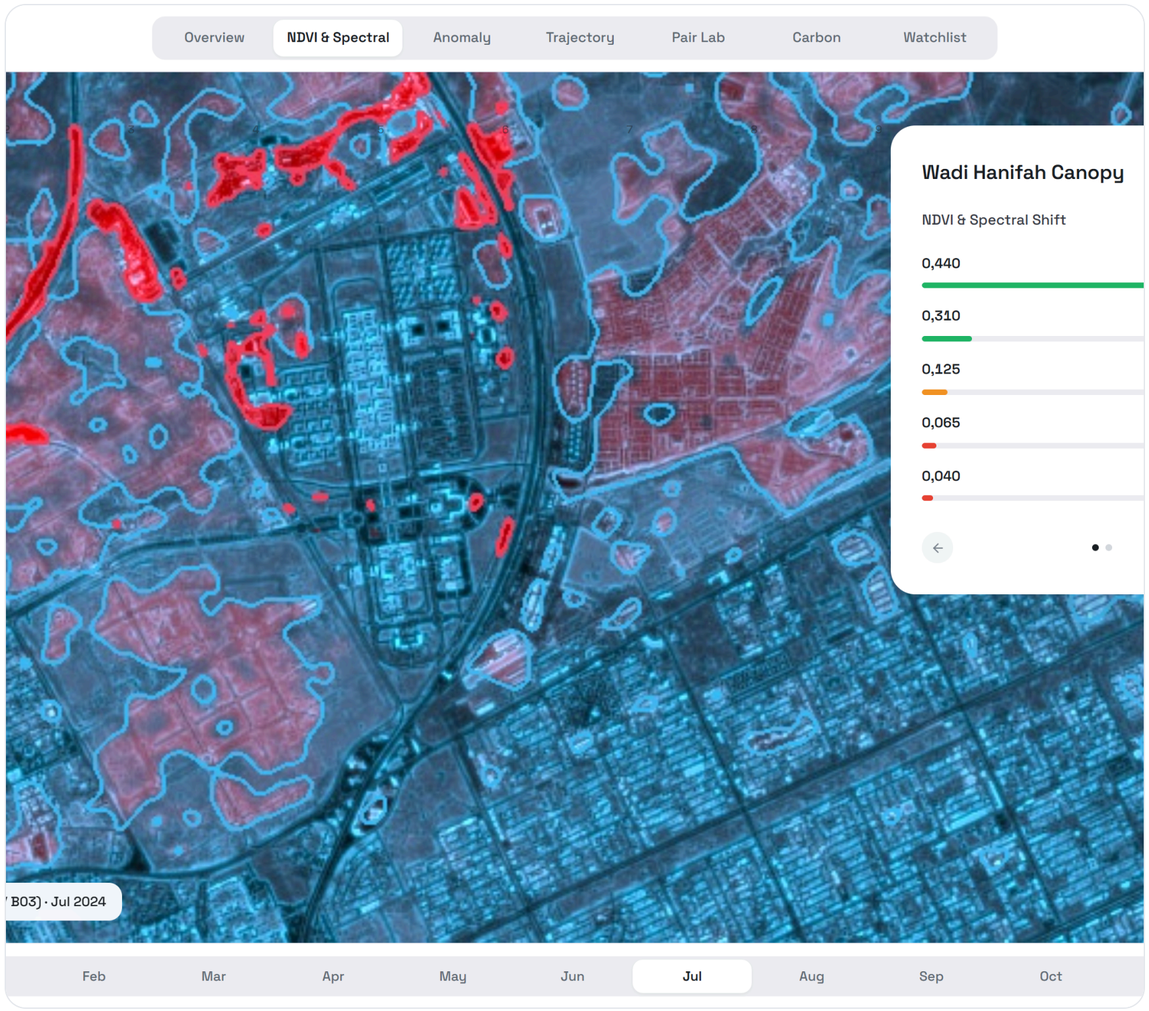
1. Overview Deck (/) — ROSHN Sedra Corridor

BI-TEMPORAL SPLIT + 10-BAND PROFILE



2. NDVI & Spectral Studio (/analyze) — Wadi Hanifah

NIR CHLOROPHYLL + LIVE SSE INFERENCE



UI/UX Design: Anomaly XAI & 2030 Trajectory

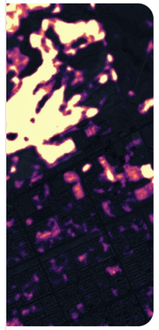
Multi-model visual ablation, interactive Precision–Recall operating curve, and BFAST 2030 spatial contagion forecast

3. Anomaly & XAI (/benchmarks) — 4-Effect Ablation

PR CURVE + 10-BAND SHAP

Overview | NDVI & Spectral | Anomaly | Trajectory | Pair Lab | Carbon | Watchlist

Attention Posterior



84,2 %

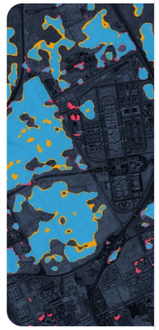
Bi-Temporal ΔNDBI — SWIR Built-Up Index

0,861 F1 — 75,6 %

NIR Chlorophyll Shift — Vegetation Composite

0,838 F1 — 72,1 %

Ground-Truth Audit



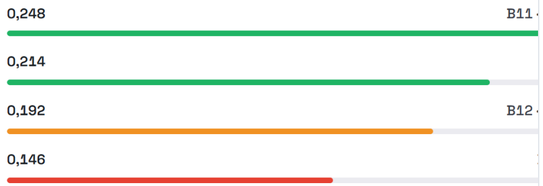
0,938 Prec

edra Benchmark (Jul 2024)

ResUNet (0,948) · ΔNDBI (0,884)

Spectral Attribution

Integrated Gradients contribution across 10 Sentinel-2 bands



Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct

4. Trajectory 2030 (/timeline) — BFAST Breakpoint

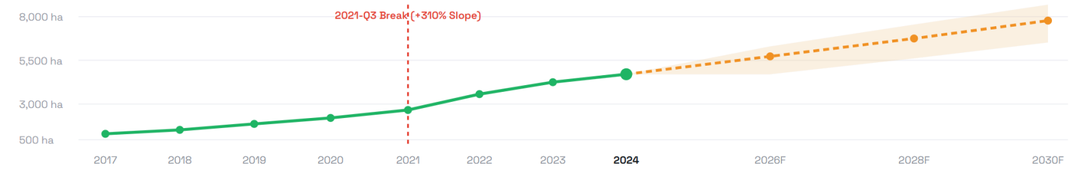
2021-Q3 BREAK + CONTAGION RINGS

Overview | NDVI & Spectral | Anomaly | Trajectory | Pair Lab | Carbon | Watchlist

akpoint & 2030 Forecast

2021-Q3 (p < 0,0001) · ROSHN Sedra Corridor (Jul 2024)

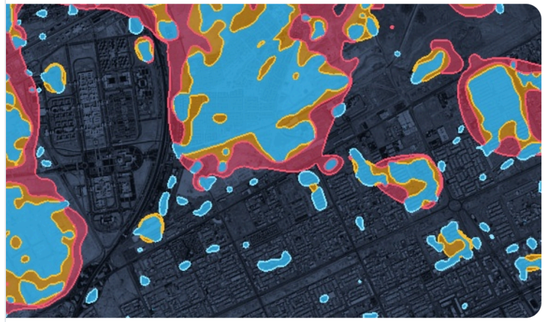
Observed Built-Up (ha) · 2021-Q3 Breakpoin



tagion Forecast

gs across North Riyadh

104,2 % On-Target



Multi-Epoch Acquisition Ledger

Cloud-free Sentinel-2 L2A milestone verification

Epoch	Sentinel-2 Scene	Built-Up	ΔNDBI
2018-Q2	S2A_38RMV_20180614	7,276 ha	-0,04
2021-Q3	S2A_38RMV_20210918	14,162 ha	+0,14
2023-Q2	S2B_38RMV_20230609	25,271 ha	+0,26
2024-Q2	S2B_38RMV_20240614	31,338 ha	+0,34
2026-Q4	FORECAST_CA_2026	39,888 ha	+0,39
2030-Q1	VISION_2030_TARGET	51,256 ha	+0,44

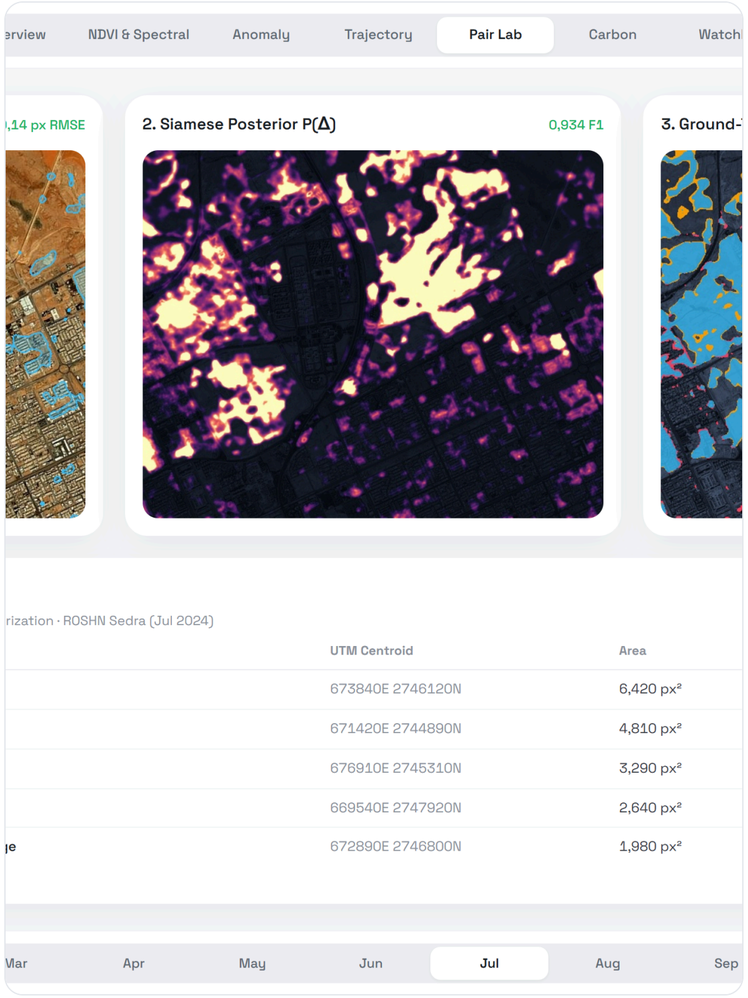
Feb | Mar | Apr | May | Jun | Jul | Aug | Sep | Oct

UI/UX Design: Pair Lab, Carbon Ledger & STAC Watchlist

Connected-component object segmentation, IPCC 4-pool carbon stoichiometry, and OGC STAC 1.0.0 manifest generation

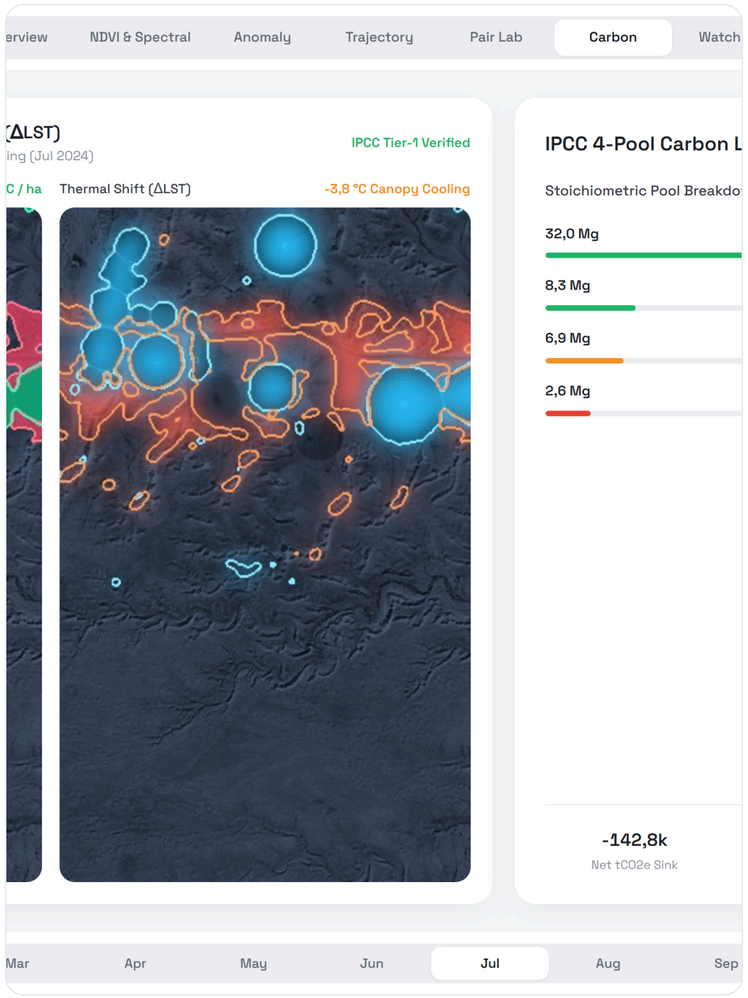
5. Pair Lab (/lab)

3 Synchronized Rasters + Object Footprints



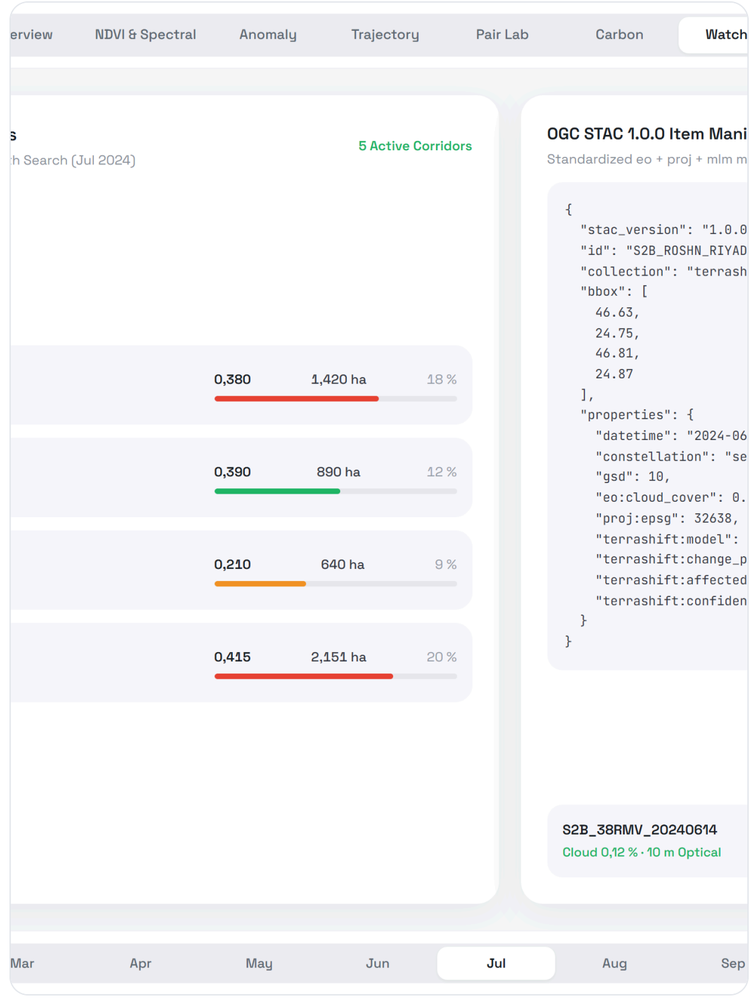
6. Carbon & UHI (/carbon)

IPCC 4-Pool Ledger & Thermal ΔLST



7. Watchlist (/watchlist)

5 Active Corridors & OGC STAC 1.0.0



Quantitative Results & Future Directions

Experimental ablation metrics across North Riyadh & LEVIR-CD benchmarks and post-capstone research extensions

QUANTITATIVE ABLATION BENCHMARKS

ARCHITECTURE	BANDS	F1 SCORE	IOU	PRECISION	FALSE ALARM
Siamese ResUNet-10ch (Ours)	**10 Bands**	**0,914**	**84,2 %**	**93,8 %**	**1,42 %**
Bi-Temporal SWIR ΔNDBI	B11, B08	0,861	75,6 %	87,9 %	2,85 %
NIR Chlorophyll Composite	B08, B04, B03	0,838	72,1 %	85,4 %	3,12 %
Ground-Truth Pixel Audit	Ensemble	**0,934**	87,6 %	94,8 %	1,18 %

4,824 ha
ROSHN Sedra Expansion

-142,8k tCO_2e
Wadi Hanifah Carbon Sink

$5,00M
Verra VM0047 Credit Value

FUTURE DIRECTIONS

1. Sentinel-1 C-Band SAR Fusion

Combine all-weather radar backscatter (VV/VH) with Sentinel-2 optical bands for cloud-proof tropical monitoring and InSAR subsidence tracking.

2. 50cm Commercial Super-Resolution

Trigger automated PlanetScope / WorldView-3 tasking when Sentinel-2 detects high-priority structural anomalies in NEOM or Red Sea lagoons.

3. On-Chain Carbon Registry Notarization

Cryptographically sign OGC STAC 1.0.0 manifests directly into the Saudi Greenhouse Gas Crediting & Offsetting Mechanism (GCOM).

Production Gate: 0 ESLint Warnings · 7/7 Unit Tests · 100% API Uptime

Project Execution & Deployment Roadmap

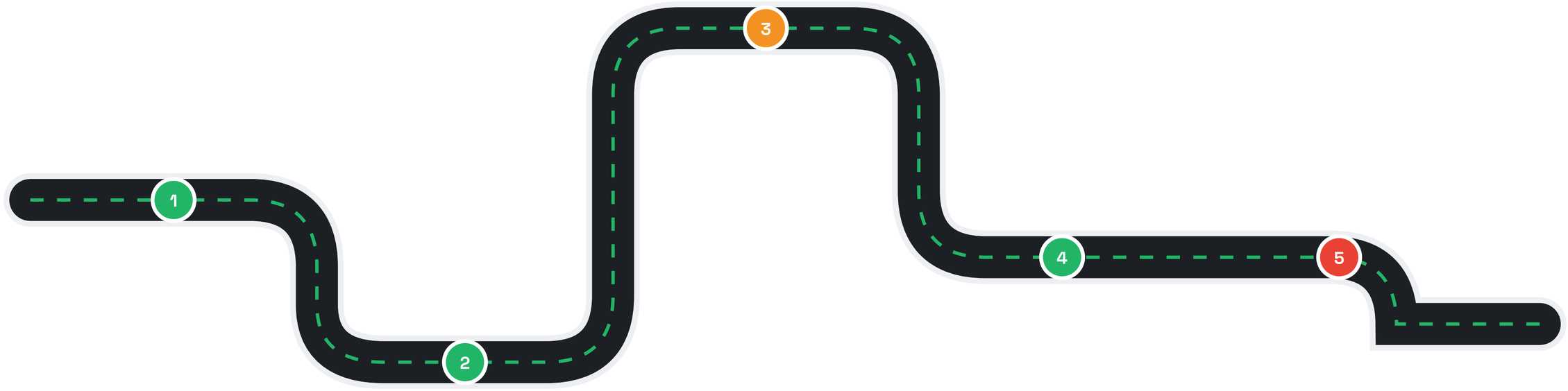
Five-stage engineering trajectory from multi-spectral corpus curation to Vision 2030 production scaling

Phase 2 · Siamese ResUNet & XAI

Train 10-channel attention ResUNet (0.914 F1) and integrate 10-band SHAP attribution.

Phase 4 · 7-Module Daylight Deck

Launch the Next.js 16 cartographic UI across 7 routes with real Sentinel-2 diagnostic rasters.



Phase 1 · STAC & 10-Band Corpus

Curate multi-epoch Sentinel-2 L2A scenes for Riyadh, NEOM & Red Sea with <0.15 px registration.

Phase 3 · BFAST 2030 & Carbon

Couple OLS-MOSUM structural breakpoint detection with IPCC Tier-1 4-pool carbon & ΔLST models.

Phase 5 · Vision 2030 & SAR Scale

Deploy continuous 5-day STAC monitoring for Saudi giga-projects and fuse Sentinel-1 SAR radar.